

REPORT ON Organisation of (mxd) memory with latches



REPORT ON

Experiment No. ...1

Title Organisation of (mxd) memory with latches

Name Ishan Dasgupta, Saankhya Saswanta, Souvik Ray Date of Performance 10/1/23, 17/1/23

Semester 4th Date of Submission 14/3/23

Branch CST Roll No 2021CSB100, 2021CSB103, 2021CSB104 Signature

Examined by [Signature] 21/3/23



REPORT ON

EXPERIMENT 1

Organization of (mxd) memory with latches

Objective: To build memory module with the features of Read/Write, Din/Dout using RS latch considering:

(a) $m=1, d=1$ (b) $m=2, d=1$

Apparatus: Logic Probe, Trainer's Kit

Components:

- a) RS latch (74279)
- b) Tri-state Logic (74367)
- c) Triple-Input NAND (7410)
- d) Inverter (7404)

Procedure:

Realize the logic circuits shown in Figure 1 and verify as per the following table:
(I and II)

TABLE I
Verification for Design (a)

S.No.	Select	R/W	Data in (supply)	Data out (verify)	Activity
1	1	0	1	-	Write 1 in Loc
2	1	1	-	1	Read 1 from Loc-1
3	0	x	-	-	No operation
4	1	0	0	-	Write 0 in Loc-1
5	1	1	-	0	Read 0 from Loc-1

NOTE: We refer to TABLES I and II only for 1x1 and 2x1 memory respectively. TABLE III which shows verification for 2x2 memory is not used. It has been provided for further explanation on larger memory module design.



REPORT ON

TABLE-II

Verification for Design (b)

S.No.	Select	R/W	Din	Dout	Verify
1	1	0	1(d ₁)	-	Write 1 in Loc-1
2	0	0	1(d ₂)	-	Write 1 in Loc-0
3	1	1	-	1(d ₁)	Read 1 from Loc-1
4	0	1	-	1(d ₂)	Read 1 from Loc-0
5	1	0	0	-	Write 0 in Loc-1
6	0	0	1	-	Write 1 in Loc-0
7	1	1	-	0	Read 0 from Loc-1
8	0	1	-	1	Read 1 from Loc-0
9	1	0	1	-	Write 1 in Loc-1
10	0	0	0	-	Write 0 in Loc-0
11	1	1	-	1	Read 1 from Loc-1
12	0	1	-	0	Read 0 from Loc-0

TABLE-III

Verification for Design (c) (2x2 memory)

S.No.	Select	R/W	D1 _{in}	DO _{in}	D1 _{out}	DO _{out}	Verify
1	1	0	1	0	-	-	Write 10 in Loc-1
2	0	0	0	1	-	-	Write 01 in Loc-0
3	1	1	-	-	1	0	Read 10 from Loc-1
4	0	1	-	-	0	1	Read 01 from Loc-0
5	1	0	0	0	-	-	Write 00 in Loc-1

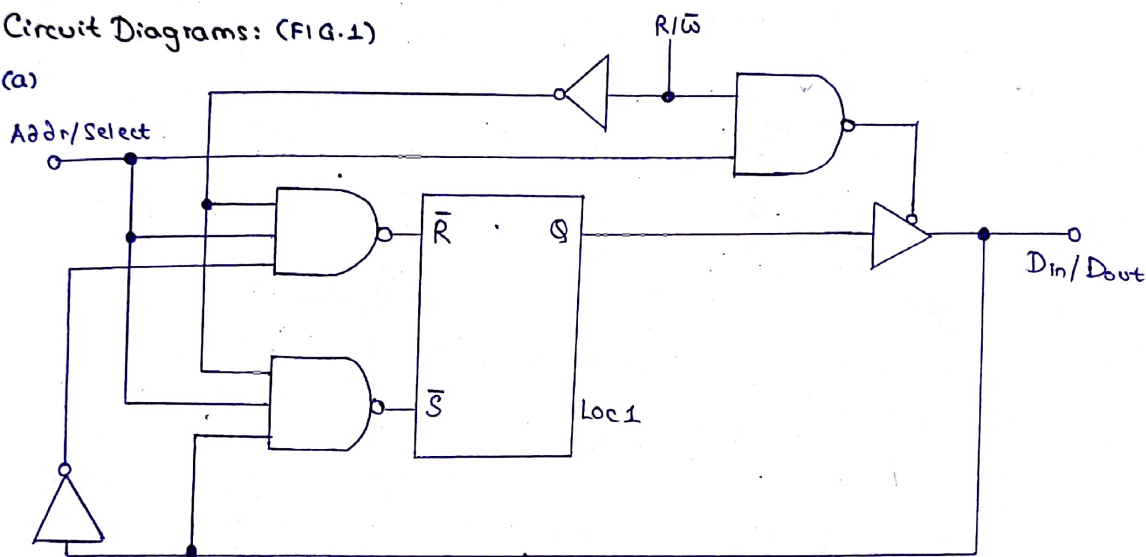


REPORT ON

S.No.	Select	R/ $\bar{W}$	D1 _{in}	D0 _{in}	D1 _{out}	D0 _{out}	Verify
6	0	0	1	1	-	-	Write 11 in Loc-0
7	1	1	-	-	0	0	Read 00 from Loc-1
8	0	1	-	-	1	1	Read 11 from Loc-0

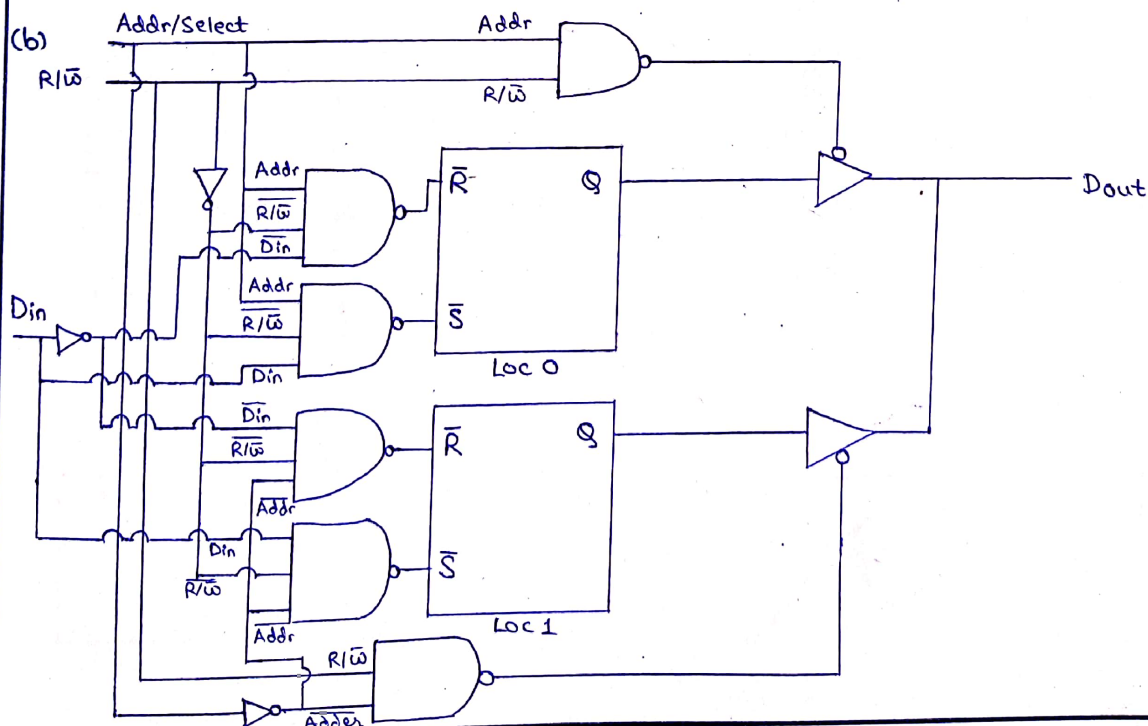
Circuit Diagrams: (FIG-1)

(a)



▲ 1-bit 1-word memory

(b)



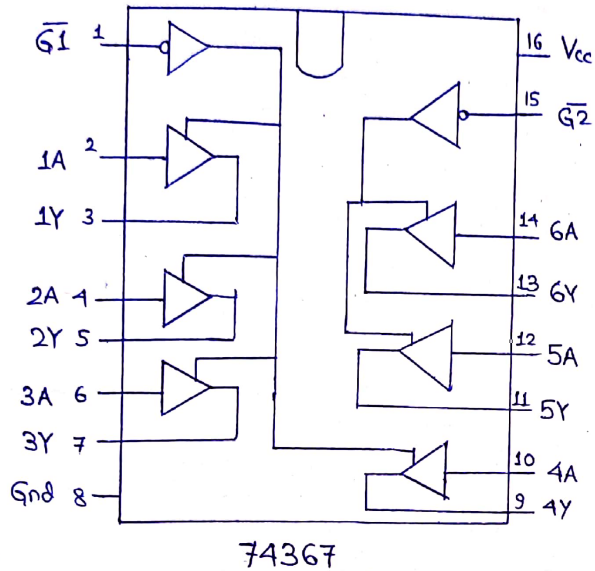
▲ 1-bit 2-word memory



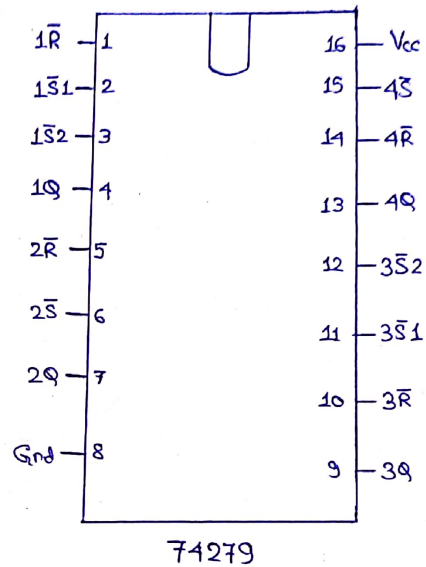
REPORT ON

Chip Configurations:

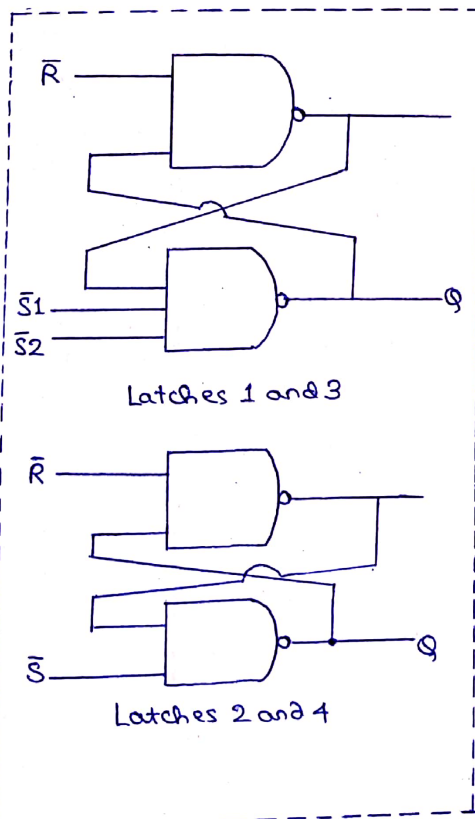
(a)



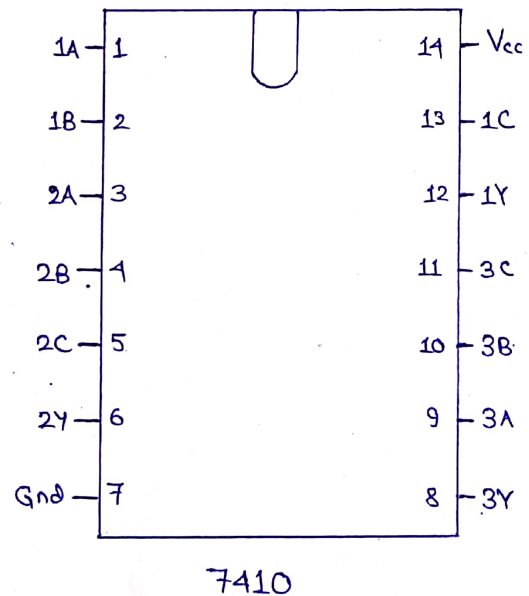
(b)



(c)



(d)





Indian Institute of Engineering Science And Technology, Shibpur

Department of Computer Science & Technology

REPORT ON

Verification Table:

(a) 2x1 memory:

Select	R/ $\bar{W}$	D _{in}	SR Latch output
1	0	1	1 → loc 0
1	0	0	0 → loc 0
0	0	0	0 → loc 1

(b) 2x1 memory:

Select	R/ $\bar{W}$	D _{in}	Expected D _{out}	Observed D _{out}
1	0	1(d ₁)	-	-
0	0	1(d ₂)	-	-
1	1	-	1(d ₁)	1
0	1	-	1(d ₂)	1
1	0	0	-	-
0	0	1	-	-
1	1	-	0	0
0	1	-	1	1
1	0	1	-	-
0	0	0	-	-
1	1	-	1	1
0	1	-	0	0

Signature.....



Indian Institute of Engineering Science And Technology, Shibpur
Department of Computer Science & Technology

REPORT ON

(C) 1X1 MEMORY

Sl.No	SELECT	R/W	Din	Expected Dout	OBSERVED Dout
1	1	0	1	-	-
2	1	1	-	1	1
3	0	x	-	-	-
4	1	0	0	-	-
5	1	1	-	0	0
6	1	0	0	-	-
7	1	0	-	-	-
8	1	1	-	0	-



Indian Institute of Engineering Science And Technology, Shibpur

Department of Computer Science & Technology

REPORT ON

Conclusion:

In this experiment we designed and verified the working of i) 1x1 memory - there is 1 word of 1 bit size, (ii) 2x1 memory - there are 2 words of 1 bit size each.

While doing the above, we also understood the working of 74367 Tri-state logic. It consists of 2 active low enables $\overline{G1}$, $\overline{G2}$. $\overline{G1}$ is connected to 4 tristate buffers and $\overline{G2}$ to 2 tristate buffers. When the enables are connected to ground (low), the tristate buffers work normally and when the enables are connected to V_{cc} (high), the tristate buffers are in high impedance state.

There are 4 lines for each memory:

- Select/Address line - It activates the memory when connected to V_{cc} /high. In case of 2x1 memory, select specifies memory location at which operation is to be performed. At select=0, Loc-0 is activated and select=1, Loc-1 is activated.
- R/ $\overline{W}$ - It specifies operation to be performed on memory. At high/1, R=1 and hence memory read operation is performed. At low/0, i.e. W=1, write operation is performed, i.e. content is written to memory.
- Din - Here input is provided to memory to be written.
- Dout - Content of memory is read here.

In case of 1x1 memory:

For:

Select	R/ $\overline{W}$	Din	Expected Dout	Observed Dout
1	0	0	-	-
1	0	-	-	-
1	1	-	0	-

In 1st row, we write 0 to memory.

In 2nd row, we write - to memory.

In 3rd row, we read content of memory.



REPORT ON

We would expect 0 to be Dout in 3rd row, (the content of memory would be expected to be 0 because in 2nd row, we wrote nothing to memory).

But we get Dout as -. This is because the value of hanging connection is -, taken to be new content of memory. Hence observed Dout is -.

In case of 2x1 memory:

We check whether correct data is written to correct location by checking whether output of corresponding SR latch is same as data supplied or not.

REPORT ON Memory Testing



REPORT ON

Experiment No. 2

Title Memory Testing

Name Ishan Dasgupta, Saankhya Samanta, Souvik Roy Date of Performance 24/1/23, 31/1/23, 7/2/23

Semester 4th Date of Submission 14/3/23

Branch CST Roll No 2021CSB100, 2021CSB103, 2021CSB104 Signature

Examined by



REPORT ON

EXPERIMENT 2

Memory Testing

Objective: To become familiar with read/write operation of the memory and verification of the data written.

Apparatus: Trainer's Kit, Logic Probe

Components:

- (a) RAM (6116)
- (b) 12-stage binary counter (CD 4040)
- (c) 4-bit comparator (7485)

Procedure:

a) Perform data write to RAM (6116) and read data from the RAM. Set address and data manually. (Refer to Figure 1).

(i) Write to RAM:

- 1. Set $\overline{OE} = H$.
- 2. Set $\overline{CS} = L$.
- 3. Set address at A_1, A_0 .
- 4. Set data $I/O_2, I/O_1$.
- 5. Set $\overline{WE} = L$.
- 6. Set $\overline{WE} = H$. Go to Step 3 for write to next location.

(ii) Read from RAM:

- 1. Set $\overline{WE} = H$ and $\overline{CS} = L$.
- 2. Change address A_1, A_0 .
- 3. Set $\overline{OE} = L$.
- 4. Read data at $I/O_2, I/O_1$.

b) (i) Perform fast data write to RAM (6116) in l locations ($l \leq m$ of an $m \times n$ memory). Generate memory address using counter (CD 4040). Consider 4-bit data. Data is input from the least significant 4 bit of the counter.

(ii) Perform data read from RAM. Compare the data read from memory and the data actually written (use comparator 7485) to verify the condition of memory.

Date

Page No.

Signature.....



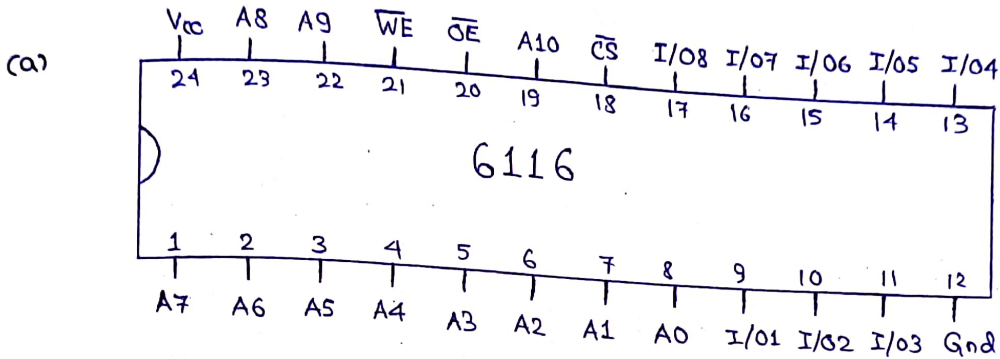


Indian Institute of Engineering Science And Technology, Shibpur

Department of Computer Science & Technology

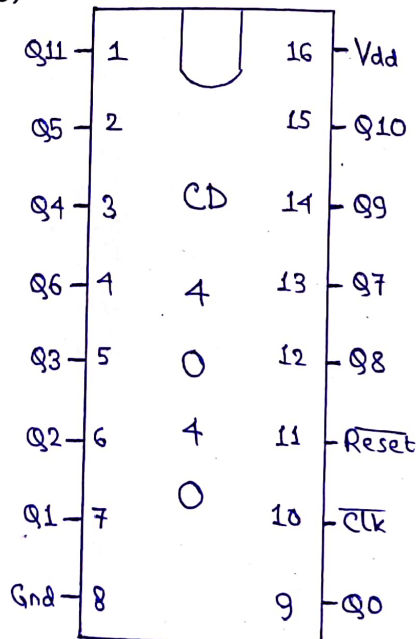
REPORT ON

Pin Configuration:

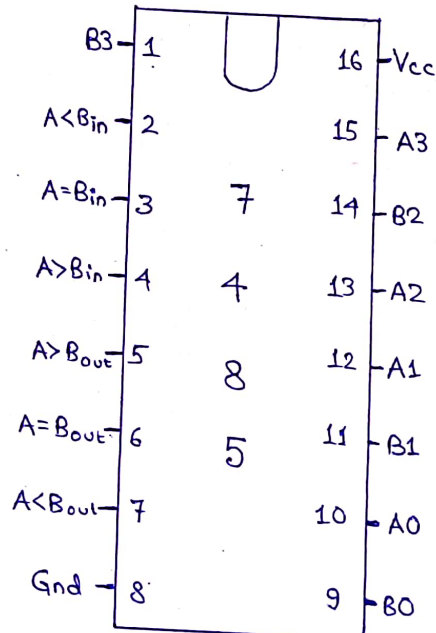


$\overline{CS}$	$\overline{WE}$	$\overline{OE}$	Event
0	0	1	Write
0	1	0	Read

(b)



(c)



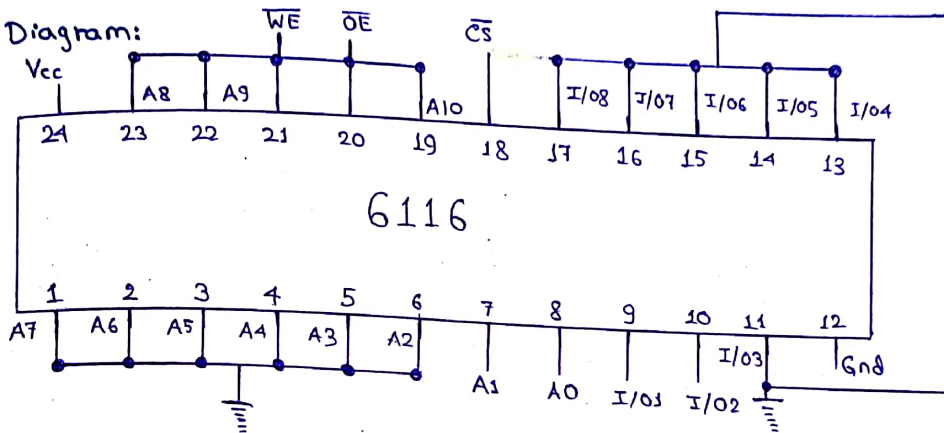
Set A = B_in to H
A > B_in to L
A < B_in to L.



REPORT ON

Observations:

1a) ■ Circuit Diagram:



▲ FIGURE 1

Memory read/write (manual)

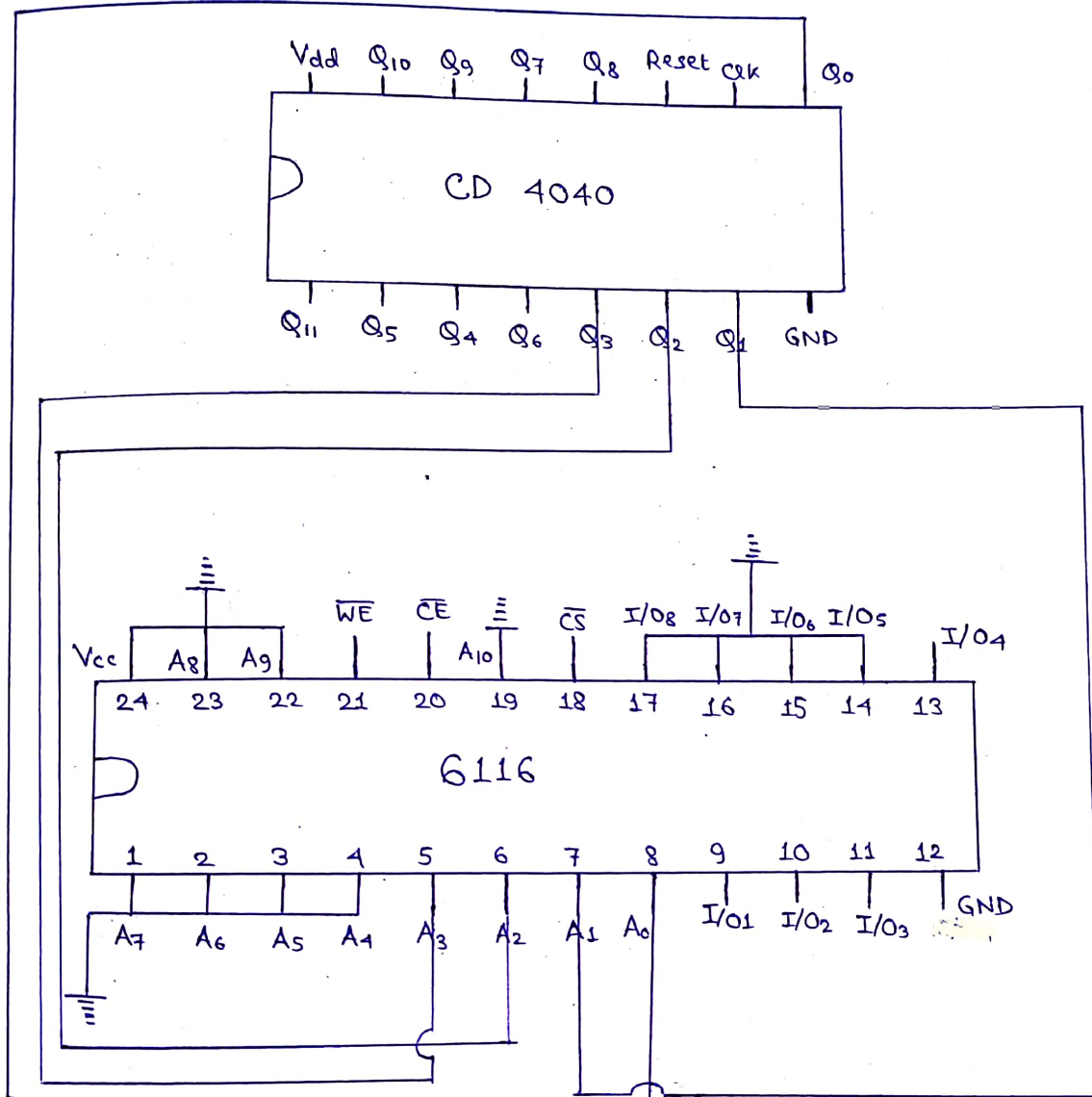
■ Truth Table:

$\overline{CS}$	$\overline{OE}$	$\overline{WE}$	A_1	A_0	I/O_2	I/O_1	Operation
0	1	0	0	0	0	0	Write 00000000 to location 0000000000
0	1	1	0	1	0	1	Write 00000001 to location 0000000001
0	1	0	0	1	0	1	Write 00000001 to location 0000000001
0	1	1	1	0	1	0	Write 00000010 to location 0000000010
0	1	0	1	0	1	0	Write 00000010 to location 0000000010
0	1	1	1	1	1	1	Write 00000011 to location 0000000011
0	1	0	1	1	1	1	Write 00000011 to location 0000000011
0	0	1	0	0	0	0	Read 00000000 from location 0000000000
0	0	1	0	1	0	1	Read 00000001 from location 0000000001
0	0	1	1	0	1	0	Read 00000010 from location 0000000010
0	0	1	1	1	1	1	Read 00000011 from location 0000000011



REPORT ON

■ Circuit Diagram:



Result:

We considered the 4 least significant bits of counter i.e., $Q_3 Q_2 Q_1 Q_0$ for setting locations in RAM.

At address $0000000Q_3Q_2Q_1Q_0$, we fast data write $0000Q_3Q_2Q_1Q_0$.

Then we read the data from the locations.



REPORT ON

Conclusion:

Here we perform i) Manual and ii) Fast Data Write and Data Read to and from memory RAM 6116. We also came to know about the chip's working.

RAM 6116 has 11 address lines - from A0 to A10, thus there are 2^{11} approx 2000 possible addressable locations and 8 data lines from I/O1 to I/O8.

The following terminals of RAM 6116 are essential:

- i) $\overline{CS}$ - Chip Select - It tells whether RAM should be enabled or not. At low, it activates RAM.
- ii) $\overline{OE}$ - Output Enable - It tells RAM 6116 whether outputs I/O1, I/O2, ..., I/O8 should give output or not. At high, the terminals I/O1, ..., I/O8 are used to take input of required address, and at low, they give outputs of required address.
- iii) $\overline{WE}$ - Write Enable - It tells RAM 6116 if it should perform write operation or not. At low, it enables RAM 6116 to take input and write it to required memory location. At high, it disables write operation.

We give data at any address $xyzw$ as $xyzw$, during data write in this experiment.

The changing of address is done manually in manual data write and data read. We ground A2 to A10 and I/O2 to I/O8. We thus vary the 2 least significant bits of the address/memory location and hence data at address 00000000A1A0 is 000000A1A0.

In case of fast data write, fast data read, we connect least 4 significant ~~beats~~ bits of address i.e. A3, A2, A1, A0, to least 4 significant bits of counter Q3, Q2, Q1, Q0 respectively. Remaining address lines and input/output lines are grounded. As the clock pulse changes value, value of counter changes and accordingly the address and hence data to be written, during fast data write also changes. During fast data read, we can also get the output at any 000000A3A2A1A0, using this.

Date

Page No.

Signature.....





REPORT ON Data Transfer between MDR and RAM



REPORT ON

Experiment No. 3

Title Data Transfer between MDR and RAM

Name Ishan Dasgupta, Saankhya Samanta, Soubik Ray Date of Performance 14/12/23, 21/12/23

Semester 4th Date of Submission 14/3/23

Branch CST Roll No 2021CSB100, 2021CSB103, 2021CSB104 Signature

Examined by



REPORT ON

Experiment Number 3

Data Transfer between MDR and RAM

Objective:

To become familiar with data transfer from buffer (MDR) to main memory (RAM) and from RAM to MDR. Further, to study the data access from external world (input device) to mm.

Apparatus:

Trainer kit, Logic Probe

Components:

- 1) Octal D Latch (74373)
- 2) Tri-state noninverting buffer (74241) - 2 nos
- 3) RAM (6116)

The Design:

- 1a) Realize the logic circuit for 1 bit and verify data transfer (i) from external source E to MDR, (ii) MDR to RAM, (iii) RAM to MDR, and (iii) RAM to MDR.



REPORT ON

Procedure:

Steps to realize data transfer

(i) From E to MDR:

0. Set data to E.

1. $OP/CTRL = L$, then $Enable (G) = H$ of the octal latch

2. $Mode = H$, then $R/\bar{W} = L$ (writes E to latch - that is MDR).

(ii) MDR to RAM:

A. Read from MDR

1. $Enable (G) = L$ of the octal latch.

2. $Mode = L$, then $R/\bar{W} = H$ (reads from output of the latch).

3. Data available at the output Q.

B. Output of MDR to RAM.

0. Data available at I/O.

1. $\bar{WE} = L$ (write enabled), then $\bar{OE} = H$ (output disabled) of RAM.

2. $\bar{WE} = H$ (output of RAM are now fixed to the value Q).

(iii) RAM to MDR:

1. $\bar{OE} = L$ (output enabled), then $\bar{WE} = H$ (write disabled).

2. $Enable (G) = H$ of latch (enables write latch / MDR).

3. $Mode = L$ (to prevent external data to input to latch), then $R/\bar{W} = L$ (writes to MDR).

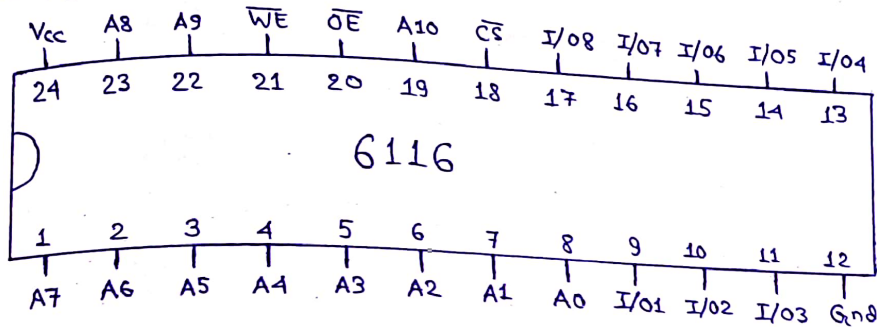


Indian Institute of Engineering Science And Technology, Shibpur

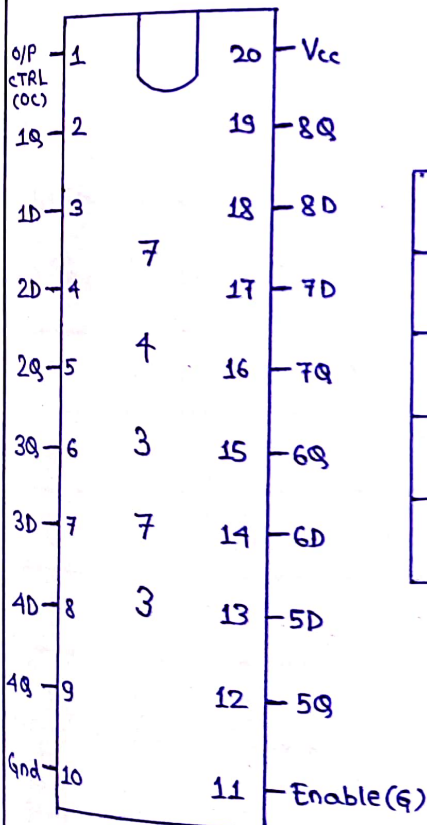
Department of Computer Science & Technology

REPORT ON

Chip Configuration:

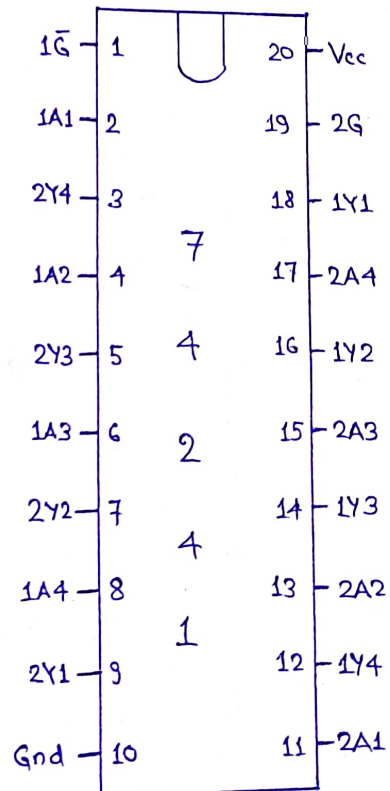


$\overline{CS}$	$\overline{WE}$	$\overline{OE}$	Event
0	0	1	Write
0	1	0	Read



O/P CTRL	G	D	Q
L	H	H	H
L	H	L	L
L	L	X	Q_0
H	X	X	Hiz

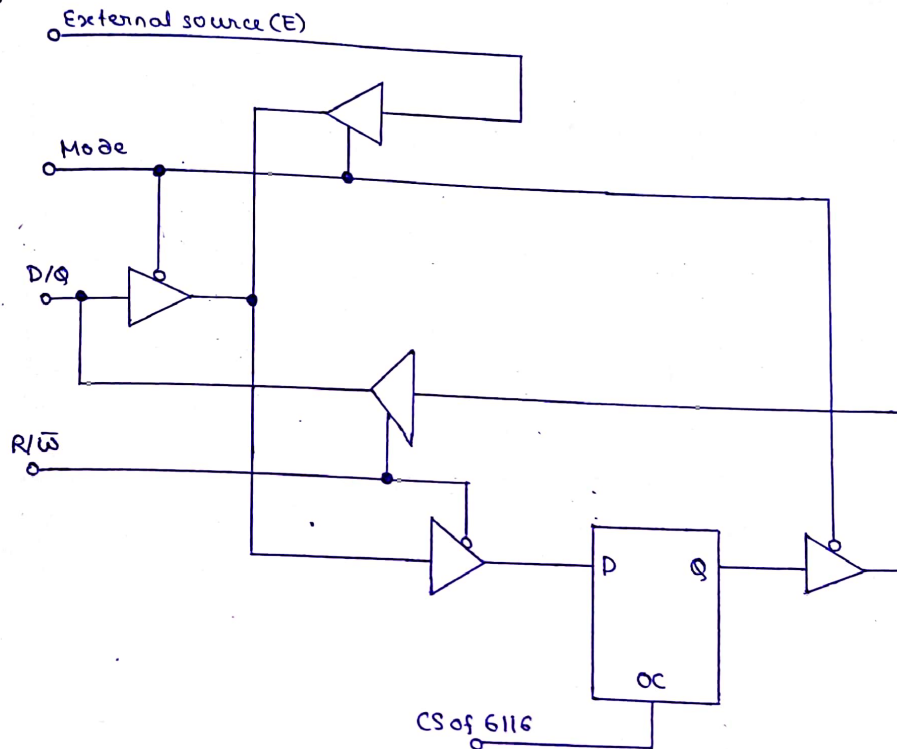
▲ For octal latch
74373





REPORT ON

Circuit Diagram:



▲ Internal structure of a block (1-bit MDR)

Observations:

ai) Truth Table:

E	OP/CTRL	G	Mode	R/ $\bar{W}$	D/Q	Operation
0	0	1	1	0	0	Write 0 to MDR from external source
0	0	0	0	1	0	Read 0 from MDR
1	0	1	1	0	1	Write 1 to MDR from external source
1	0	0	0	1	1	Read 1 from MDR

Date

Page No.

Signature.....





REPORT ON

1a ii) Truth Table:

E	OP/CTRL	G	Mode	R/ $\bar{W}$	D/Q	$\overline{CS}$	$\overline{OE}$	$\overline{WE}$	A ₀	I/O ₁	Operation
0	0	1	1	0	0	-	-	-	-	-	Write 00000000 to location 0000000000 from MDR
0	0	0	0	1	0	0	1	0	0	0	
0	0	1	1	0	0	0	0	1	0	0	
1	0	1	1	0	0	-	-	-	-	-	Read 00000000 from 0000000000 RAM
1	0	0	0	1	0	0	1	0	1	1	Write 00000001 to location 0000000001 from MDR
1	0	1	1	0	0	0	0	1	1	1	
1	0	1	1	0	0	0	0	1	1	1	Read 00000001 from 0000000001 RAM

1a iii) Truth Table:

OP/CTRL	G	Mode	R/ $\bar{W}$	D/Q	$\overline{CS}$	$\overline{OE}$	$\overline{WE}$	A ₀	I/O ₁	Operation
-	-	-	-	-	0	1	0	0	0	Write 0 to MDR from RAM
0	1	0	0	0	0	0	1	0	0	
0	0	0	1	0	0	0	1	0	0	Read from MDR
-	-	-	-	-	0	1	0	1	1	Write 1 to MDR from RAM
0	1	0	0	1	0	0	1	1	1	
0	0	0	1	1	0	0	1	1	1	Read from MDR



REPORT ON

Conclusion:

We know that the Data Register (DR) transfers data to and from memory. This is what we verify in this experiment.

In this experiment we come to know about 74373 Octal D Latch IC also. It has two main terminals:

- (i) Enable G — Active High Enable — When it is high D Latch starts to operate.
- (ii) O/P CTRL (Output Control) — Active High — When it is high, the output of D Latches present in IC will be corresponding/depending on the input of respective latches. At low it disables all outputs.

In our Data Register (DR) there are five terminals:

- (i) E — With this we can give input to data register from an external source.
- (ii) D/Q — With this we can give content of DR as input (D) and get the content of DR as output (Q).



REPORT ON

- (iii) O/P CTRL (Output Control) - It tells whether output of DR is enabled or not. At high D/A gives output Q and at low it takes input D.
- (iv) Mode - It specifies from where DR is supposed to take input. At high it takes input from external source E, at low it takes input from memory.
- (v) R/W - It specifies operation to be performed. At high, $R=1$, read operation is done which implies transfer from DR. At low ($w=1$), write operation is done which implies transfer to DR.

We use two 74241 - Non inverting buffers - it consists of two enables - one active high and another active low. There are three tristate gates for each enable. We connect enables of one buffer to mode and the enables of other to R/W.



REPORT ON

In a(i) Data Transfer from External Source E to MDR:

We write to DR from external source E (it can be 0/1). To verify correct data is written we read content of DR.

In a(ii) Data Transfer from MDR to RAM:

Through external source E we write 0/1 to MDR. The D/Q terminal of MDR is connected to A_0 and I/O , of RAM 6116 during data write. So content of RAM 6116's address 0000000000 A_0 is A_0 where content of MDR is A_0 . Now to verify whether correct information is written or not we read content of RAM 6116 at appropriate location where data was written.

In a(iii) Data Transfer from RAM to MDR:

We first write some data at a location of RAM 6116. Then we read this data and transfer this data to the MDR. To verify if correct data was written or not we read content of MDR.